

Survey of Dimensionality Reduction Techniques and their Applications for Classifying Disease State in Human RNA-seq Data

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Introduction

Machine learning can answer biological questions

- TEC-LncMir: identifies RNA-RNA interactions [1]
- Random Forest: can identify TFBS with UV damage [2]
- AlphaFold: predicts protein structure [3]

Introduction

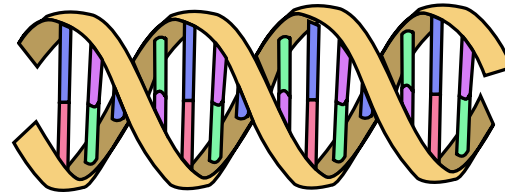
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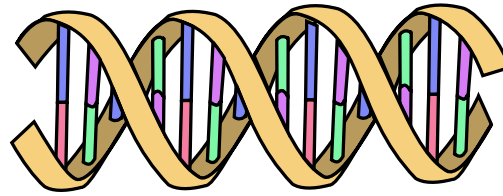
We're interested in gene expression data

- Informs how the cell is behaving
- Captures disease state, cell type, etc.
- Many diseases (cancers) result from aberrant gene expression

DNA (genes)

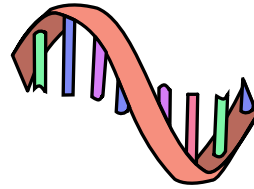


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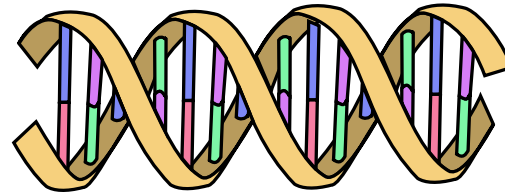


Transcription

mRNA

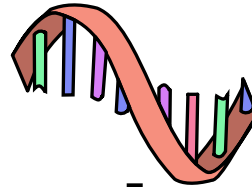


DNA (genes)



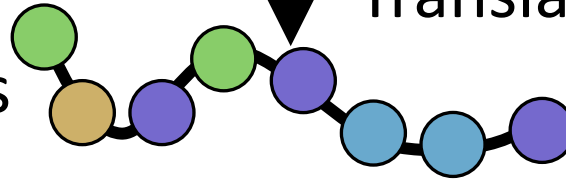
Transcription

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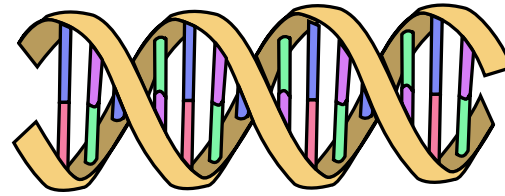


Translation

Amino Acids

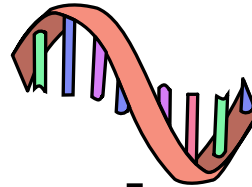


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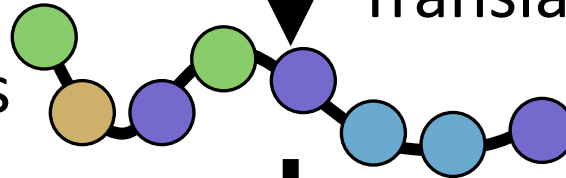
Transcription

mRNA



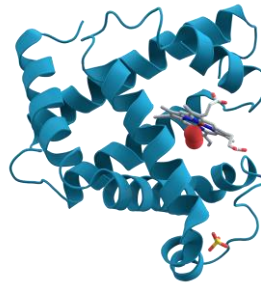
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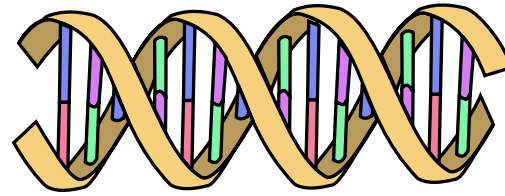


Folding

Protein

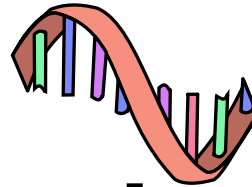


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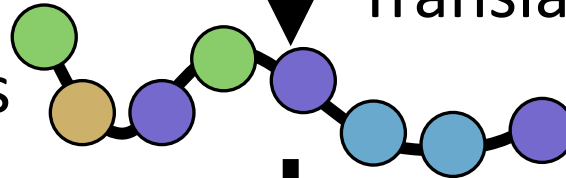
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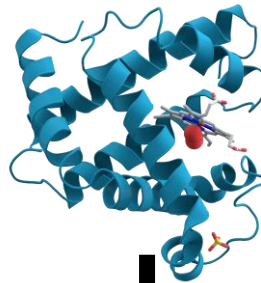
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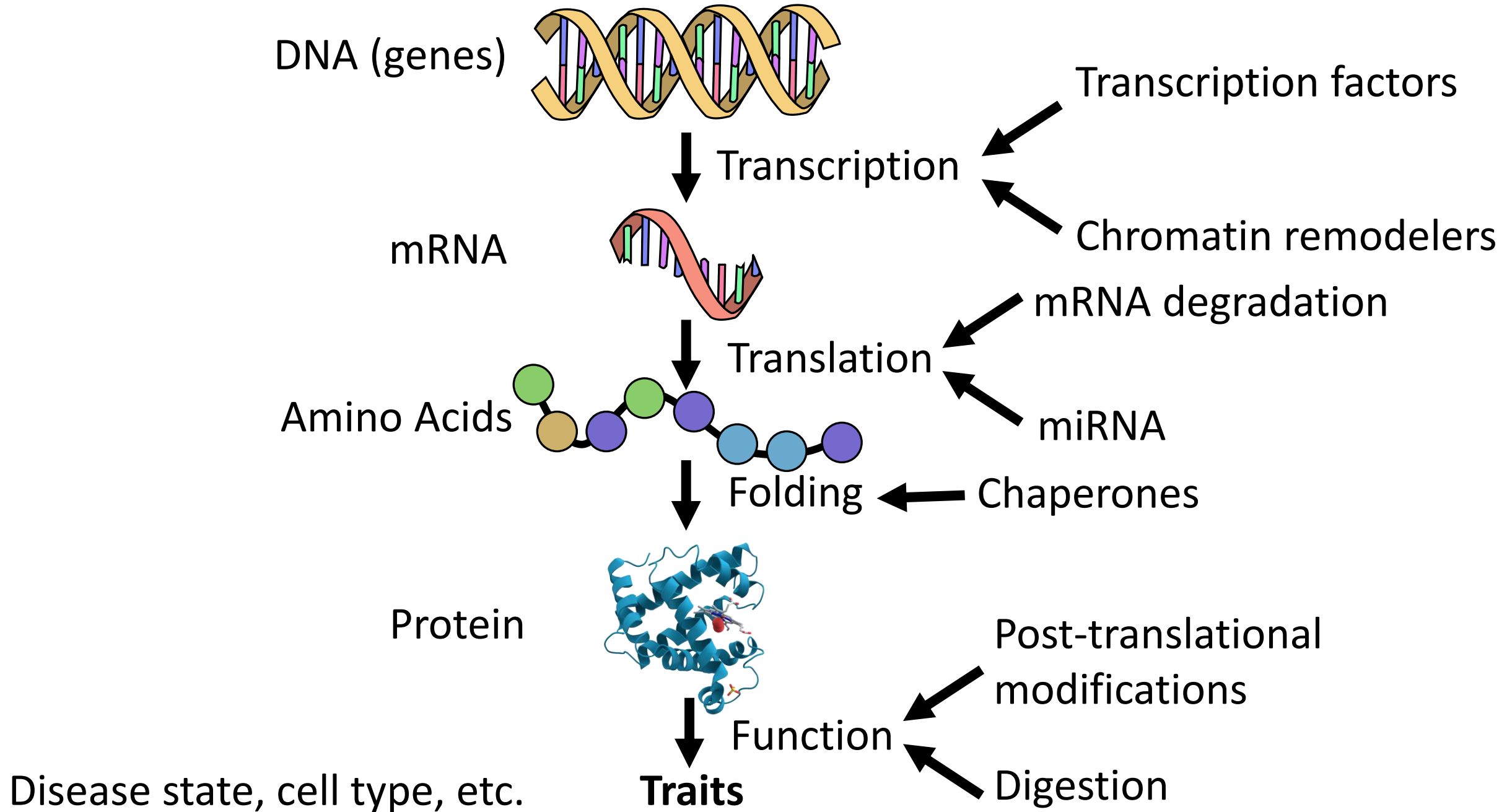
Protein



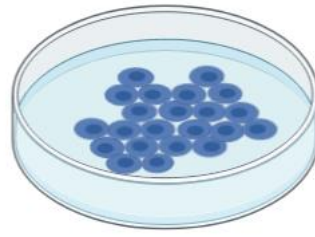
Function

Disease state, cell type, etc.

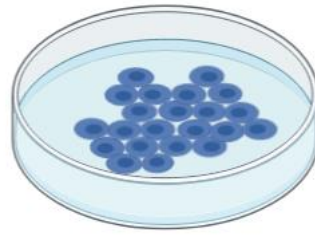
Traits



Cells



Cells

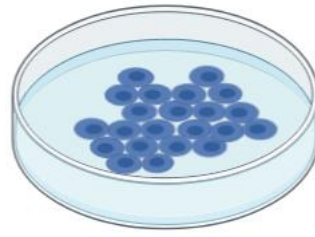


RNA isolation

Transcripts



Cells



RNA isolation



Transcripts

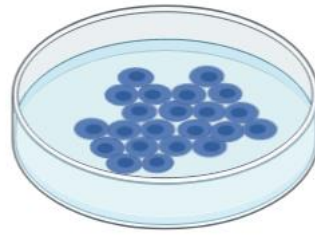


Sequencing

Sequences

AAAUUCCUUGGAAGA . . .
AAUACAGAUUCAUAG . . .
AAUCGAUAUCGACUG . . .
GAUAGUACGUACGAU . . .
CUAGAUUCUAGAUGUA . . .

Cells



RNA isolation



Transcripts



Sequencing



Sequences

AAAUCCUUGGAAGA...
AAUACAGAUUCAUAG...
AAUCGAUAUCGACUG...
GAUAGUACGUACGAU...
CUAGAUCAUGAUGUA...

Alignment



Genes

MYCN, MAPK, ERK, KRAS...

RNA-seq Data

Gene Expression Omnibus

- <https://www.ncbi.nlm.nih.gov/sites/GDSbrowser>

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Very low sample sizes

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- Inconsistent gene sets
- Variations in methodology
- Different normalization

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Challenging to learn on

- Lots of data out there
- Untapped potential

Pretrained Models?

No suitable pretrained models exist for RNA-seq data

- Most pretrained models utilize scRNA-seq data
- The few that use RNA-seq data are unsuitable

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Modern deep learning methods require vast quantities of data

- Go to the single-cell level
- Utilize massive, coordinated data collection effort

Older, simpler models are still commonplace

Survey of Dimensionality Reduction

4 dimensionality reductions :

- *a priori* aggregation
- PCA
- Kernelized PCA (kPCA)
- Nonnegative matrix factorization

5 machine learning models:

- Decision Tree (DT)
- k -Nearest Neighbors (k -NN)
- Naïve Bayes (NB)
- Random Forest (RF)
- Support Vector Machine (SVM)

a priori Aggregation

Genes are assigned categories and annotations

- Gene Ontology (GO) categorization [4,5]
- Component, function, and process

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Dimensionality reduction was done by taking the mean

- Features of reduced dataset correspond to gene groups
- Reduced dimension from $\approx 10^4 \rightarrow 10^3$

Retains biological context

PCA, kPCA, and NMF

Reduce dimension to $d = 2, \dots, 9$

- $d = 2$ allows for visualization
- $d = 9$ is still small compared to the number of data points

PCA and kPCA re-center the data, while NMF does not

All three methods lose biological tractability

- Reduced dimensions have no biological meaning

Sample	GEO Accession	Title	<i>n</i>	<i>d</i>	<i>k</i>	Ref.
UCC	GDS1615	Ulcerative colitis and Crohn's disease comparison: peripheral blood mononuclear cells	127	22283	3	[6]
SCLC	GDS2373	Squamous cell lung carcinomas	130	22283	6	[7]
SEC	GDS2771	Large airway epithelial cells from cigarette smokers with suspect lung cancer	192	22283	3	[8]
MDS	GDS3795	Myelodysplastic syndrome: CD34+ hematopoietic stem cells	200	54675	2	[9]
ALL	GDS4206	Pediatric acute leukemia patients with early relapse: white blood cells	197	54675	3	[10]
HIV	GDS4228	HIV infection and Antiretroviral Therapy effects on mitochondria in various tissues	166	4825	2	[11]
JIA	GDS4267	Systemic juvenile idiopathic arthritis and non-systemic JIA subtypes: peripheral blood mononuclear cells	154	54675	3	[12]
GBM	GDS5205	Long-term adult survivors of glioblastoma: primary tumors	70	54675	3	[13]
MDG	GDS963	Macular degeneration and dermal fibroblast response to sublethal oxidative stress	36	12625	2	[14]

Final Notes

All reported accuracy values are from leave-one-out validation

- Small dataset sizes make traditional cross-validation infeasible

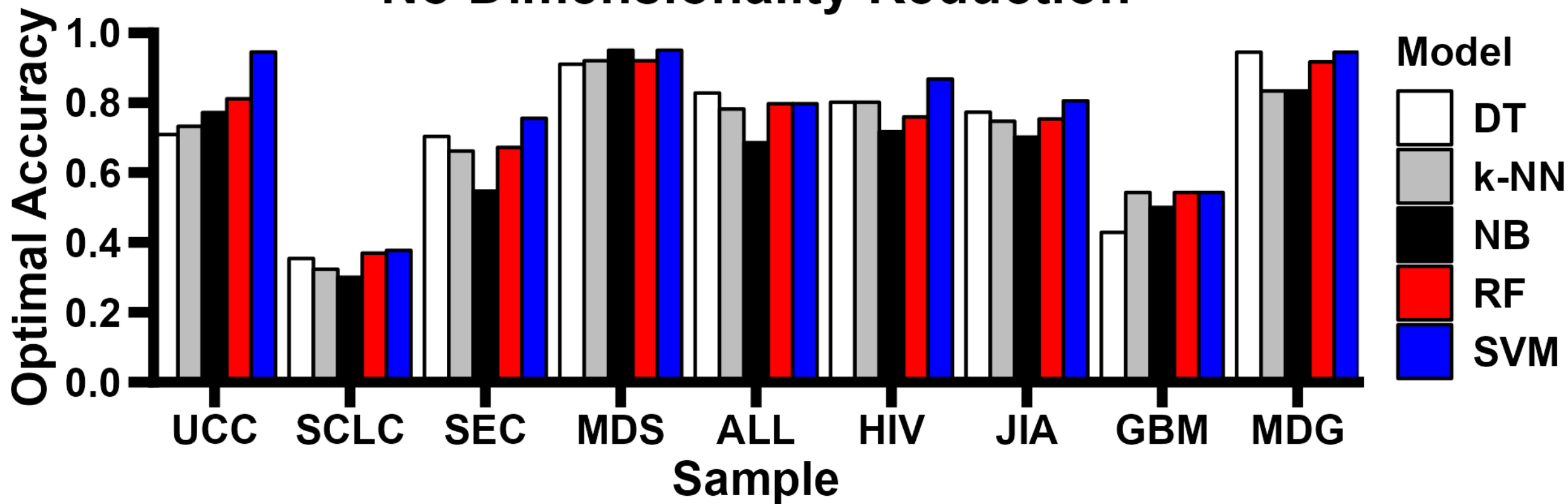
We tested a large combination of model hyperparameters

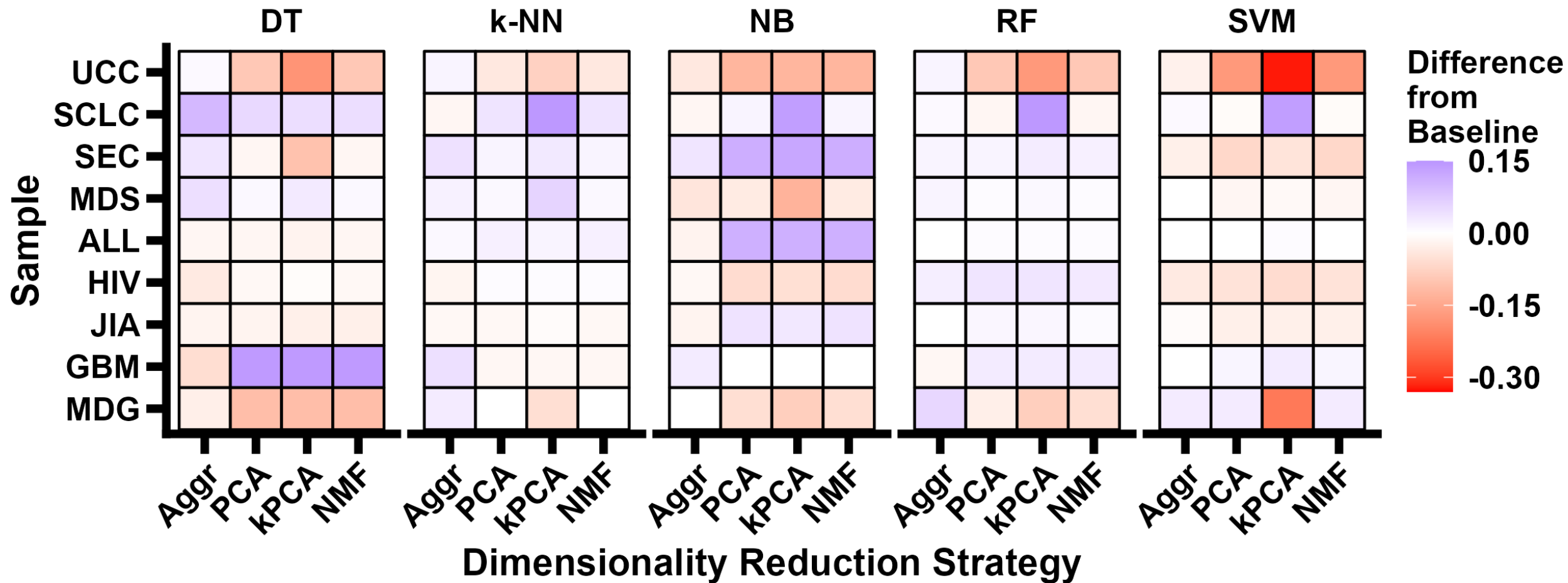
- “Optimal accuracy” is the best accuracy across all hyperparameters

All data and code is available on GitHub:

- https://github.com/nolan-middleton/CPT_S-Group-Project-Fall-2025
- Supplemental figures and supporting information can be found here

No Dimensionality Reduction





Conclusions

The impacts of dimensionality reduction are varied

- Can help or harm classification performance
- Highly variable across datasets, dimension, and classification model

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Depends on underlying biology and model behavior

- Don't need dimensionality reduction on models like SVM
- PCA, kPCA, and NMF were roughly equivalent
- PCA, kPCA, and NMF can significantly boost simple models like NB
- *a priori* aggregation may be well-suited for cancer data

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